TLSTMF-YOLO: Transfer Learning and Feature Fusion Network for Earthquake-Induced Landslide Detection in Remote Sensing Images

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***Abstract*— Detecting earthquake-induced landslides in remote sensing images is challenging due to the varying sizes of landslides, uneven distribution, and the prevalence of small targets. This study proposes a novel approach, the TLSTMF-YOLO model, which combines a C3-Swin-Transformer and multiscale feature fusion techniques to enhance detection accuracy and efficiency. Key innovations include the use of a convolutional block attention module (CBAM) to improve feature representation, and a bidirectional feature pyramid network (BiFPN) for optimized cross-scale feature fusion. To address data scarcity, a transfer learning strategy is applied, supported by an AdamW optimizer and cosine learning rate strategy for faster convergence. Evaluations on the Jiuzhaigou and Luding landslide datasets demonstrate the model's effectiveness, achieving precision, recall, and mean average precision (mAP)@0.5 of 95.7%, 89.9%, and 90.5% on the Jiuzhaigou dataset, and 96.0%, 90.9%, and 94.5% on the Luding dataset, respectively. In addition, the model processes frames efficiently, with times of 6.61 and 12.2 ms on the two datasets. These results confirm the model's capability for accurate and efficient landslide detection, highlighting its potential for real-world applications.**

***Index Terms*— Attention mechanism, landslide, multiscale feature fusion, small sample, target detection, transfer learning.**

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I. Introduction

EARTHQUAKE-INDUCED landslides are a severe secondary geological hazard characterized by their high frequency, large scale, widespread distribution, and significant damage [1], [2]. The Mw6.5 earthquake in Jiuzhaigou, China, on August 8, 2017, triggered numerous landslides and secondary disasters, resulting in 19 deaths and 343 injuries. Similarly, the MS6.8 earthquake in Luding County, China, on September 5, 2022, caused 93 deaths, 25 missing persons, and over 400 injuries, with over 80% of casualties attributed to landslides. Enhancing early identification of earthquake-induced landslides is crucial for improving risk mitigation capabilities. However, rapid and accurate identification of these earthquake-induced landslides presents significant challenges due to the varying scales of targets, uneven distribution, and dense packing of small targets [3], [4].

Traditional post-earthquake landslide emergency investigations mainly rely on manual field surveys and visual interpretation of remote sensing data [5], [6], [7]. Manual field surveys are often limited by disrupted transportation and rugged terrain, which make it time-consuming and labor intensive to quickly and effectively assess and evaluate widespread landslides. In addition, the visual interpretation of remote sensing data has limited efficiency in identifying earthquake-induced landslides, making it challenging to achieve rapid large-scale recognition [8]. These limitations affect landslide disaster assessment and hinder the effectiveness of emergency investigation efforts.

With the rapid development of artificial intelligence (AI) and computer technology, deep learning-based object detection algorithms [9], [10], [11], especially convolutional neural networks (CNNs), have garnered widespread attention [12], [13]. These algorithms can be trained on graphics processing units (GPUs) to independently learn robust object features, significantly enhancing detection accuracy over traditional methods. While expanding the receptive field and enhancing

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semantic representation, deep networks reduce the resolution of earthquake-induced landslide images through convolution operations, leading to blurred details. Conversely, shallow networks better capture geometric details but provide weaker semantic features. Early region-CNNs (R-CNNs) predicted directly from the final feature map of the neural network, lacking fine-grained spatial features, which makes small target detection challenging [14]. The single feature map output of standard CNNs fails to meet the detection requirements for earthquake-induced landslide remote sensing images and cannot effectively represent multiscale objects [15], [16].

Many works [17], [18], [19] recently have been proposed based on multiscale fusion models for identifying landslides. However, conflicts between high accuracy, computational capability, and small sample size hinder the development of target detection algorithms in earthquake-induced landslide remote sensing images. Current methods often focus on constructing deep and complex neural networks to improve detection accuracy by increasing model complexity while overlooking operational efficiency in practical engineering applications [20]. High accuracy requires models with strong feature extraction capabilities, but computational limitations restrict the practical application of such complex models. Models are prone to overfitting in small sample scenarios, affecting detection effectiveness. Transfer learning techniques can effectively utilize pretrained models under small sample conditions to enhance detection accuracy [21], [22]. However, the diverse characteristics of remote sensing images, including scale variation, small targets, large quantities, high density, and complex imaging backgrounds, increase the difficulty of algorithm design. Therefore, achieving a balance between high accuracy, computational capability, and small sample size remains challenging.

To address these challenges, this article introduces a deep transfer learning network, TLSTMF-YOLO, designed to enhance landslide detection in remote sensing images. The proposed model integrates the C3-Swin-Transformer for robust feature extraction, an attention mechanism to emphasize critical regions, the AdamW optimizer to improve convergence efficiency, and a cosine learning rate strategy to optimize model training. These innovations collectively enhance feature fusion and significantly improve prediction accuracy. In addition, the adoption of a transfer learning strategy mitigates the limitations posed by small sample sizes, enabling the network to generalize effectively across diverse datasets.

1) Proposing a feature extraction network module based on the C3-Swin-Transformer, enhancing feature extraction capabilities to meet the computational demands of high accuracy for earthquake-induced landslide remote sensing targets.
2) Designing an efficient multiscale feature fusion network module based on an attention mechanism to achieve nonuniform target context information fusion. AdamW optimizer and cosine learning rate strategies are utilized to avoid overfitting and accelerate the convergence of the network.
3) Utilizing transfer learning to address the small sample problem, enabling rapid adaptation to new tasks, improving learning efficiency and accuracy, and reducing time and resource requirements, thereby benefiting the rapid identification of earthquake-induced landslide tasks.

II. Related Works

Deep learning methods can automatically extract features from input landslide remote sensing images and classify these images. Wang et al. [17] used a residual-connected autoencoder with a multiscale structure to adapt to different scale variations. Li et al. [18] employed multiple parallel region of interest (ROI) pooling modules to map features to different scales and used a multilevel cascade method for detection. Based on Faster R-CNN, Qin et al. [19] introduced different candidate region extraction networks on different feature maps to independently detect targets of various scales. Rahman et al. [4] proposed the scale-normalized image pyramid (SNIP) training strategy to tackle the challenges of small target datasets and improve detection accuracy across scales.

Due to the limitations of single-scale features, multiscale feature fusion has gained increasing attention in multiscale object detection in earthquake-induced landslide remote sensing images. Li et al. [18] proposed a multifeature fusion network (MFFSP), which combines multiscale and multilevel contextual information to improve the detection accuracy of landslides of various sizes. Wang et al. [23] introduced a feature enhancement and fusion method based on multiscale feature mapping, enhancing the detection accuracy of landslides of small objects. Ghorbanzadeh et al. [24] proposed the extended feature pyramid network (EFPN), which enhances the landslide detection performance of small objects by adding high-resolution pyramid feature layers. Wei et al. [25] designed a novel detector that improves landslide detection of small objects by utilizing object features and contextual information. However, these approaches still face challenges in dealing with complex terrain and multiscale features, which can affect the model's robustness and accuracy in real-world applications. Thus, there is a need for more effective methods that can better capture multiscale information and adapt to diverse environmental conditions. Liu et al. [26] proposed a detection method based on feature layer fusion, expanding convolutional networks to learn lower level positional information, reducing feature channels via convolution on mid-level features, and then fusing the features. Although these multiscale feature fusion methods effectively improve small object detection performance, they often lose the interrelationship between different scale features during the fusion process, limiting further performance enhancement.

In addition, attention mechanism-based methods are also effective for multiscale feature fusion. By enhancing or suppressing relevant and irrelevant information and balancing the importance of different scale features, these methods can improve the overall discriminative capability of the features. Yu et al. [3] proposed a hierarchical-attention multiscale deconvolution network (HADeenNet), which refines low-level features through various attention modules, filters out background noise, and adaptively integrates spatial and semantic

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information of intermediate-level features, significantly boosting the landslide detection performance of small objects. Zhang et al. [27] introduced a small object attention network that improves landslide detection of small objects by adding an attention mechanism to shallow feature layers, thereby reducing unnecessary background information. Su and Chai [28] incorporated two types of attention modules into the feature pyramid network (FPN), representing semantic relevance in both the channel and spatial dimensions, thus preventing the landslide feature information of small objects from being overwhelmed.

The images used for landslide recognition come from various sources with significant resolution and geographical location differences. Currently, different models exhibit varied performances across different research areas and datasets [29]. There is an urgent need for a model with higher generalization performance that is capable of adapting to landslide images from different sources while achieving stable and excellent recognition accuracy. Remote sensing image target detection requires high precision while considering operational efficiency in practical applications. Target detection algorithms based on deep learning typically rely on building deep neural networks and utilizing large amounts of data to improve detection accuracy [30]. However, landslide samples are often scarce within research areas, which can lead to overfitting in machine learning models. This occurs because the model may memorize the limited examples rather than learning generalizable features, resulting in poor performance on unseen data. As a result, the model may exhibit low recognition accuracy and fail to detect landslides in diverse environmental conditions or new datasets, ultimately affecting its robustness and reliability. In scenarios with large samples, manual labeling of a significant number of samples for training is costly and lacks practical benefits for emergency response, hindering the realization of rapid intelligent earthquake-induced landslide recognition.

Transfer learning technology is essential for retraining existing models to adapt to new tasks. Its core idea is to utilize knowledge from pretrained models to reduce the need for large-scale datasets [21], [31]. Typical transfer learning methods include fine-tuning, feature extraction, freezing layers, and multitask learning [22], [32]. Research on transfer learning mostly focuses on image classification and natural language processing tasks, with relatively fewer applications in object detection tasks. This is because transfer learning models require adaptive adjustments when facing different tasks, making direct application to object detection more impractical. Without effective adjustments and optimizations, transfer learning may not perform satisfactorily in object detection tasks.

Achieving high accuracy with a small sample size is critical for the practical application of earthquake-induced landslide remote sensing image target detection. Further research is warranted on how to construct a high-precision remote sensing target detection network under small sample conditions. Asadi et al. [33] proposed a deep transfer learning model adapted to identify landslides, which effectively improved detection accuracy by utilizing pretrained models and a small number of samples for retraining. Jiang et al. [34] introduced transfer learning mask R-CNN (TL-Mask R-CNN), a remote sensing change detection network based on small sample learning, significantly enhancing detection performance under small sample conditions through transfer learning and data augmentation techniques. Bhuyan et al. [35] proposed a transfer learning strategy, which effectively improved the performance of small sample landslide target detection models without increasing annotation costs.

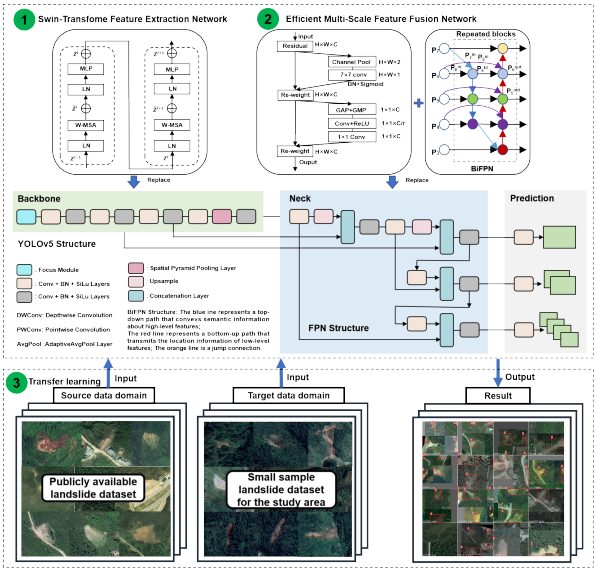


Fig. 1. Optimization techniques used in STAFM-YOLO.

III. Proposed Method

The proposed method optimizes and enhances the YOLOv5-large network, resulting in an efficient network named TLSTMF-YOLO, as shown in Fig. 1. First, to reduce model parameters and computational load, a new backbone is formed utilizing the C3-Swin-Transformer. Second, convolutional block attention module (CBAM) modules are integrated into the neck layers to enhance the network's ability to accurately localize and recognize objects of interest, thus avoiding significant computational overhead. The FPN structure is replaced by a bidirectional FPN (BiFPN) structure, accompanied by a more effective multiscale fusion method to improve accuracy. Furthermore, the network is trained using AdamW optimizer and cosine learning rate strategy, accelerating model convergence, and enhancing training speed. In addition, transfer learning enables feature and parameter sharing, reducing training costs and improving results. This technique also addresses the issue of small sample sizes. This procedure is explained in detail in Sections III-A–III-D below.

A. C3-Swin-Transformer Feature Extraction Network

Traditional CNN-based methods for segmenting landslide shapes in remote sensing images often suffer from significant information loss due to pooling operations in multiple convolutional layers [36]. To address this issue, Dual Transformers introduce a novel hierarchical structure by integrating the shifted window self-attention mechanism (Fig. 2), dividing the

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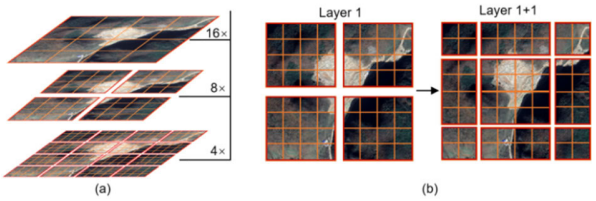


Fig. 2. Hierarchical representation (a) sliding window and (b) Swin-Transformer.

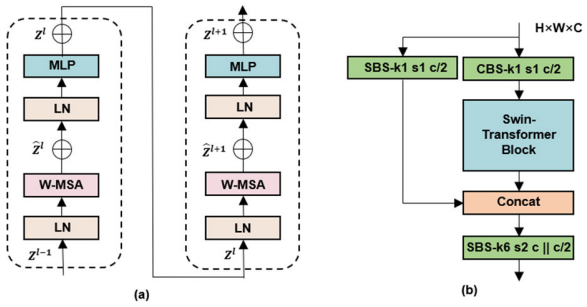


Fig. 3. Swin-Transformer block and C3STR block. (a) Adds layer normalization for stability and uses residual connections to enhance local feature capture. (b) Integrates dual Transformers into C3 to improve contextual understanding.

input image into several small patches, and facilitating information exchange among them [37], [38]. Unlike traditional self-attention mechanisms, the shifted window self-attention calculates similarities only between each element and its neighboring elements within a specific range, thus avoiding excessive similarity computations and improving computational efficiency. The Dual Transformer block comprises three components: window multihead self-attention, shifted window multilayer self-attention, and multilayer perceptron [see Fig. 3(a)]. Layer normalization is inserted in between to enhance training stability, and residual connections are applied after each module. The Transformer encoder block enhances the ability to capture different local information [39] and can be represented as follows:

$$\hat{X}^{l} = W - \mathrm{MSA}\left(\mathrm{LN}\left(X^{l-1}\right)\right) + X^{l-1} \tag{1}$$

$$X^{l} = \mathrm{MLP}\left(\mathrm{LN}\left(\hat{X}^{l}\right)\right) + \hat{X}^{l} \tag{2}$$

$$\hat{X}^{l+1} = \mathrm{SW} - \mathrm{MSA}\left(\mathrm{LN}\left(X^{l}\right)\right) + X^{l} \tag{3}$$

$$X^{l+1} = \mathrm{MLP}\left(\mathrm{LN}\left(\hat{X}^{l+1}\right)\right) + \hat{X}^{l+1}. \tag{4}$$

The C3 module serves as a crucial component of the YOLOv5 network, aiming to increase the network's depth and receptive field, thereby enhancing its feature extraction capability. However, augmenting the receptive field simultaneously may lead to acquiring irrelevant information. To ignore irrelevant information and focus solely on essential data to improve the transferability of learned features, this article proposes integrating Dual Transformers into the C3 module, forming the C3STR module [see Fig. 3(b)], to enhance the C3 module's ability to capture contextual relationships.

B. Multiscale Feature Fusion Network

Landslide detection in remote sensing images is challenging due to significant variations in target sizes, numerous small targets, and high detection difficulty. To address these issues, we propose a simplified yet effective multiscale feature fusion network that utilizes the CBAM [40] and BiFPN [41]. These techniques enhance detection accuracy while maintaining a compact model.

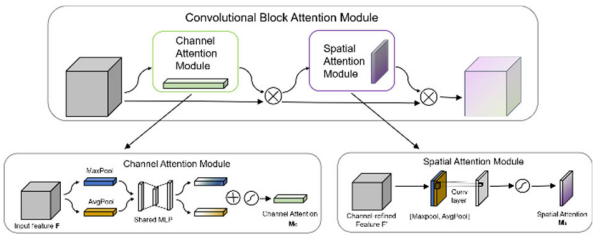


Fig. 4. Schematic of CBAM attention module.

The CBAM improves feature representation by incorporating both channel and spatial attention mechanisms. The channel attention module (CAM) prioritizes important features, while the spatial attention module (SAM) refines feature localization, helping the model better detect landslides of different sizes. By focusing on both channel and spatial attention, CBAM provides a more comprehensive feature representation, as shown in Fig. 4.

The original YOLOv5 structure employs FPN [42] to directly merge features in the feature fusion process. The BiFPN plays a crucial role in improving multiscale feature fusion, which is particularly important for landslide detection due to the wide range of target sizes in remote sensing images. Traditional FPNs suffer from the problem of unequal contributions from features of different resolutions. In contrast, BiFPN uses a bidirectional fusion strategy, which allows for a more balanced integration of features from multiple scales. This is achieved through both horizontal and vertical information transfer, making full use of multiscale features.

Furthermore, BiFPN employs an adaptive feature weighting mechanism that assigns different importance levels to each feature map. This enables the network to prioritize more relevant features and reduce the influence of less informative features. This adaptive mechanism is critical in improving detection accuracy, as it ensures that the most useful features contribute more significantly to the final output.

The effectiveness of BiFPN is demonstrated through its cross-layer connections, which facilitate efficient feature fusion by shortening information transfer paths between upper and lower layers. This improves the overall fusion process, enabling the model to generate more accurate and reliable predictions. As shown in Fig. 5, BiFPN effectively reduces unnecessary connections between feature nodes, further improving computational efficiency.

C. AdamW Optimizer and Cosine Learning Rate Strategy

In YOLOv5, stochastic gradient descent (SGD) is frequently employed. SGD can mistakenly select data points during updates, causing significant iteration errors. Furthermore, SGD may become trapped in local minima and fail to utilize all dataset samples per update [43]. In contrast, the Adam optimizer is suitable for large datasets and high-dimensional

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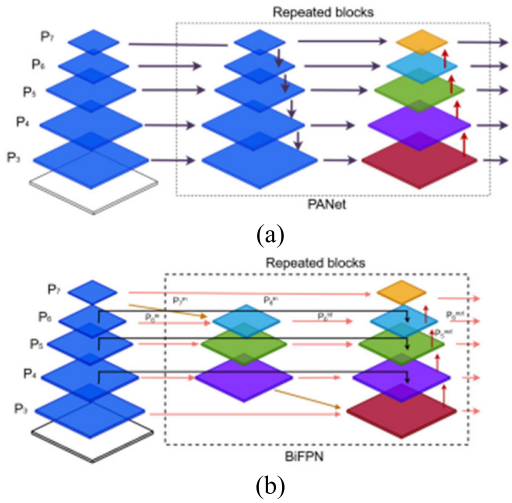


Fig. 5. PANet and BiFPN structure. (a) PANet. (b) BiFPN.

spaces with lower memory requirements. Given its advantages, this study uses an improved version, the AdamW optimizer [44], for training the image description task. The AdamW optimizer effectively addresses Adam's weight decay issue, leading to a more stable optimization process. Compared to Adam, AdamW adds a regularization term

$$R = -\alpha * (\lambda * \omega) \tag{5}$$

where λ represents the weight decay factor, usually set to 0.01 or 0.005. Finally, this study uses cross-entropy loss as the loss function for the optimizer

$$H(p,q) = -\sum_{i=1}^{m} p(x_i)\log(q(x_i)) \tag{6}$$

where m represents the number of classes, p_i represents the true value, and q_i represents the predicted value.

Adjusting the learning rate is crucial in deep learning model training. The learning rate optimizes the gradient of the loss function by adjusting network weights. A small learning rate results in slow loss function changes, extending convergence time. While a low learning rate ensures no local minima are missed, it also increases convergence time. To adapt to the training process, the learning rate should gradually decay with training batches for smooth convergence to the minimum. This study employs a cosine annealing strategy to adjust the learning rate, using a higher initial rate for fast approximation to the minimum and a lower rate later for fine-tuning the model.

D. Transfer Learning

Transfer learning, a popular machine learning technique, accelerates adaptation to new tasks by leveraging previous learning experiences [45], [46]. However, directly training landslide detection models on visual object classification (VOC) datasets can result in slower convergence and reduced accuracy due to domain differences. Utilizing transfer learning to leverage shared feature spaces between domains can help mitigate these challenges. In this study, as shown in Fig. 6,

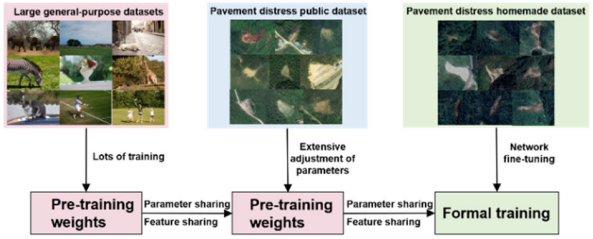


Fig. 6. Transfer learning process.

we adopted a transfer learning approach with two source domain information. Initially, the initial model was trained on the larger VOC dataset to provide optimized parameters for landslide classification. Then, extensive parameter adjustments were made on the publicly available Bijie landslide dataset, utilizing pretrained weights for the test dataset, thereby transferring the feature extraction capability of the backbone network. Despite differences in datasets, their target features are universal. This method facilitates sharing features and parameters, reducing training costs, and improving training effectiveness. Specifically, during the transfer learning process on the landslide dataset, the following parameter adjustments were made.

1) Learning Rate Adjustment: In the transfer learning process, the learning rate was gradually adjusted to avoid overfitting or underfitting issues caused by parameter updates being too fast or too slow during training on the new dataset. A higher initial learning rate helps the model quickly adapt to the new task, while gradually decreasing the learning rate ensures fine-tuning of the model.

2) Batch Size Adjustment: Batch sizes were adjusted to adapt to the new dataset's characteristics and the model's memory requirements. Appropriate batch sizes can balance the stability and efficiency of model training, aiding in better feature extraction.

3) Optimizer Adjustment: Parameters of the Adam optimizer were selected and adjusted based on the new task's characteristics to ensure that the optimization process is stable and efficient. These adjustments help the model converge better during training, improving the final detection performance.

IV. Datasets and Settings

The study evaluated the performance of the proposed algorithm through model detection using a public landslide remote sensing dataset and two custom-built datasets. Comparison with classical object detection algorithms validated the superior performance of the proposed method.

A. Datasets

The proposed model was trained and tested on the publicly available Bijie landslide dataset to validate the model's effectiveness and ensure accurate landslide detection and classification results [47]. A sample of 770 landslides was cropped from TripleSat satellite imagery taken between May and August 2018. Black triangles indicate actual landslide locations. Landslide samples vary in size. However, the publicly available Bijie landslide dataset has some limitations,

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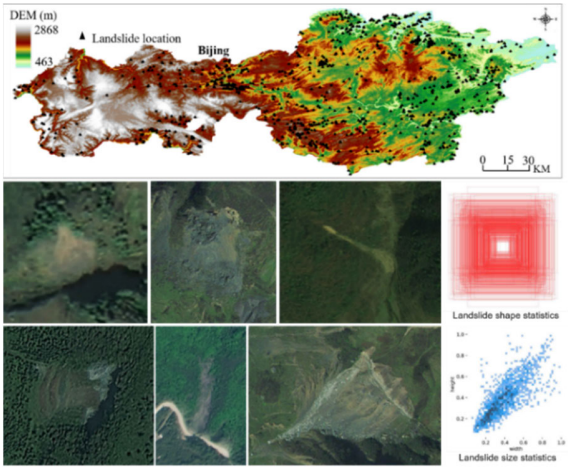


Fig. 7. Image of the Bijie landslide dataset.

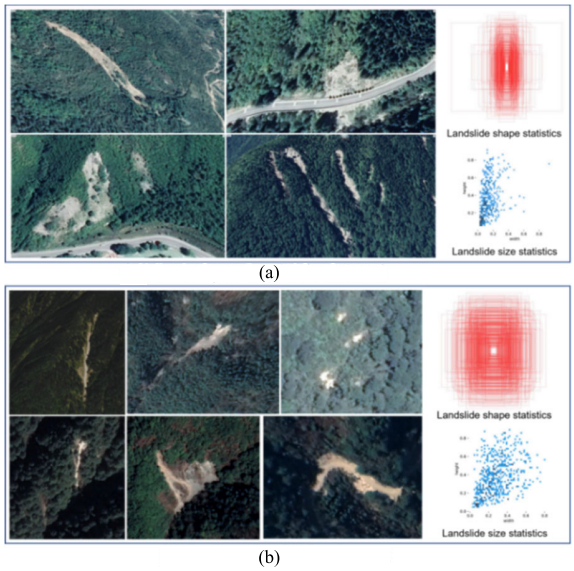


Fig. 8. Image of the customized landslide dataset. (a) Jiuzhaigou landslide dataset. (b) Luding landslide dataset.

as its spatial resolution is lower. To address this issue, visually ambiguous landslide listings were removed by experienced experts, as illustrated in Fig. 7. The shapes and sizes of landslides vary significantly among different images.

To objectively evaluate the effectiveness and robustness of transfer learning methods in practical applications, it is necessary to assess remote sensing images from different environmental backgrounds and establish high-quality evaluation datasets. Two custom datasets were created using remote sensing images from two areas that experienced landslides due to recent earthquakes (the Jiuzhaigou earthquake in 2017 and the Luding earthquake in 2022): the Jiuzhaigou landslide dataset and the Luding landslide dataset, detailed in Fig. 8. The geographic location of the landslide was determined based on optical remote sensing images of the landslide and data from the geohazards department. Based on the location of the landslide, satellite images of the landslide were extracted from Google Earth. The Jiuzhaigou landslide dataset comprises 391 images, while the Luding landslide dataset comprises 503 images, both with a resolution of 8192×5080. Landslides in these datasets are mainly distributed within complex background objects, with background objects exhibiting unique spectral and textural features across different images. Landslide color and texture features vary significantly among different images, and landslides exhibit different distribution patterns and geometric shapes. Before conducting experiments using these datasets, image labeling was performed using LabelImg software. To ensure the accuracy and consistency of the labels, we implemented a rigorous quality control process, where each labeled image was reviewed by multiple experts in remote sensing and landslide detection. Any discrepancies or ambiguities were resolved through iterative feedback, ensuring high-quality, reliable labels for the training and evaluation of the model.

B. Data Augmentation

To enhance the training effectiveness of deep learning networks, ensuring an ample quantity and high quality of data to cover various landslide scenarios is imperative. However, in practice, gathering a dataset that fully encapsulates all conceivable circumstances of landslides is often challenging. To address this limitation, data augmentation techniques are employed to process the raw dataset [48]. These techniques incorporate a variety of operations, including geometric and color transformations, increasing the dataset size to 1667 samples.

Geometric transformations involve random horizontal and vertical translations, rotations, scaling, cropping, and flipping. Meanwhile, random variations in hue, saturation, and brightness are applied in color transformations to generate additional samples, simulating diverse scenarios encountered in the real world. Through dataset augmentation, the effective number of training samples is significantly increased. This approach allows for better utilization of limited resources, enabling the simulation of a broader range of situations and enhancing the model's accuracy in recognizing various conditions.

C. Experiment Settings

STAFM-YOLO is an end-to-end network implemented in Python that is optimized for advanced computer vision tasks. The network operates on a server equipped with Ubuntu 18.04, powered by two Intel Xeon Gold 6248 CPUs, which collectively provide 80 logical CPUs and 503 GB of RAM, supporting intensive data processing requirements. In addition, the system includes two NVIDIA Tesla V100-PCIE-32-GB GPUs, each offering 32 GB of VRAM, which is crucial for handling complex graphical computations. Throughout the training process, all parameters are segregated into independent training units, encompassing convolutional kernel weights and bias terms. Training parameters were carefully optimized. All parameters, including convolutional kernel weights and bias terms, were divided into independent training units. The training utilized a batch size of 32 and the AdamW optimizer with a learning rate of 0.001. The optimizer was configured with a momentum of 0.74747, a weight decay of 0.00025, and a learning rate scheduler transitioning from an initial

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learning rate of 0.0033 to a final multiplier (lrf) of 0.15184. A warmup phase spanning approximately 3.43 epochs was implemented, with warmup parameters, including a momentum of 0.59032 and a bias learning rate of 0.18742. The model was trained for 200 epochs in total. In addition, the input image size is adjusted to 640 × 640. During data loading, CPU utilization is optimized by employing eight workers.

D. Model Assessment

The following evaluation metrics were considered to assess the model's performance in terms of detection speed, accuracy, and parameter count: precision, recall, mean average precision (mAP), and frames per second (FPS). FPS denotes the number of images that the model can process in one second, quantifying the operational speed of the model. Intersection over Union (IoU) on the joint network is a crucial evaluation metric that describes the proximity of predicted results to ground truth. It is represented as

$$\text{IoU} = \frac{\text{Area}(B_p \cap B_g)}{\text{Area}(B_p \cup B_g)} \tag{7}$$

where B_p represents the predicted bounding box and B_g represents the ground truth bounding box.

An IoU threshold of 0.5 is set to determine the accuracy of predictions. This study utilized precision, recall, and mAP to evaluate recognition performance. Precision measures the accuracy of target predictions and is calculated as the ratio of true positives (TPs) to the total number of positive predictions (TP + FP) [49]. It can be expressed as

$$\text{Precision}(\%) = \frac{\text{TP}}{\text{TP} = \text{FP}} \times 100. \tag{8}$$

Recall evaluates whether all actual targets have been detected, reflecting the proportion of TPs to the total number of actual targets. It is defined as

$$\text{Recall}(\%) = \frac{\text{TP}}{\text{TP} = \text{FN}} \times 100. \tag{9}$$

mAP is a key indicator for assessing detection accuracy in object detection tasks. AP is the average precision obtained by integrating precision–recall (PR) curves for each class. mAP is calculated by averaging the AP values for all classes. It is defined as

$$\text{mAP} = \int_0^1 \text{P(R)dR}. \tag{10}$$

V. Experimental Analysis and Results

A. Ablation Study

To evaluate the detection performance of the STAFM-YOLO algorithm, ablation experiments were conducted to assess the impact of each improvement (Swin-Transformer, CBAM, BiFPN, cosine learning rate, AdamW optimizer, and transfer learning). YOLOv5 served as the baseline model, and a check mark "√" indicated the inclusion of each method. The performance metrics included precision, recall, and mAP@50.

Table I presents the experimental results of STAFM-YOLO on the Jiuzhaigou landslide dataset. The introduction of the Swin-Transformer significantly improved the model's performance, increasing precision by 17.2%, recall by 22.5%, and mAP@50 by 22.2%. Adding CBAM increased precision by 13.6%, raised recall from 63.0% to 75.9%, and improved mAP@50 by 23.8%. Combining Swin-Transformer, CBAM, and BiFPN, the feature extraction capability was strengthened, leading to a 17.9% increase in precision, a 4.1% increase in recall, and a 3.1% rise in mAP@50, from 67.0% to 87.9%.

TABLE I
Results of Ablation Experiments on the Jiuzhaigou Dataset

Baseling	ST	CBAM	BiFPN	CTR	AdamW	TL	P (%)	R (%)	mAP50@(%)
YOLOv5	-	-	-	-	-	-	75.4	63.0	67.0
	√	-	-	-	-	-	88.4	77.2	81.9
	-	√	-	-	-	-	85.7	75.9	83.0
	-	√	√	-	-	-	82.8	79.7	82.1
	√	-	√	-	-	-	84.9	78.5	82.0
	√	√	-	-	-	-	89.6	75.4	84.1
	√	√	√	-	-	-	88.9	85.9	87.4
	√	√	√	√	-	-	89.1	87.4	88.7
	√	√	√	-	√	-	92.3	89.1	87.9
	√	√	√	√	√	√	95.7	89.9	90.5

NOTE: ST=Swin-Transformer; CTR=Cosine learning rate; TL=Transfer Learning; P=Precision; R=Recall.

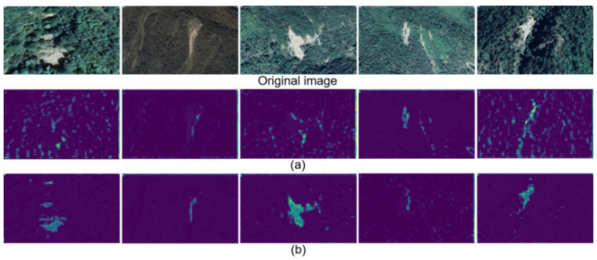


Fig. 9. Heatmaps of YOLOv5. (a) STAFM-YOLO and (b) Jiuzhaigou landslide dataset.

Fig. 9 illustrates the heatmap comparison between YOLOv5 and STAFM-YOLO on the Jiuzhaigou landslide dataset. STAFM-YOLO exhibits a higher focus on target areas and less interference in nontarget regions than YOLOv5. Particularly, in detecting various-sized targets, STAFM-YOLO efficiently concentrates on positive sample areas, thereby reducing irrelevant environmental disturbances. This indicates that integrating the CBAM mechanism enhances the network's feature representation capability. In contrast, the utilization of the BiFPN structure significantly improves the model's performance in multiscale remote sensing target detection. The inclusion of the cosine learning rate added 0.2% to precision, 1.7% to recall, and 1.4% to mAP@50. Replacing the optimizer with the AdamW optimizer further reduced network parameters, increasing precision by 3.8%, recall by 3.7%, and mAP@50 by 0.6%. This demonstrated the performance advantages of the cosine learning rate and AdamW optimizer. In addition, introducing transfer learning to reduce the randomness of initial model weights, in combination with Swin-Transformer, multiscale feature fusion, cosine learning rate, and AdamW optimizer, improved precision by 26.9%, recall by 42.7%, and mAP@50 by 35.1%. The experimental results confirmed the effectiveness of each improvement.

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TABLE II
RESULTS OF ABLATION EXPERIMENTS ON THE LUDING DATASET

Baseling	ST	CBAM	BiFPN	CTR	AdamW	TL	P (%)	R (%)	mAP50 @(%)
YOLOv5	-	-	-	-	-	-	75.4	70.4	69.3
	√	-	-	-	-	-	83.9	85.8	86.4
	-	√	-	-	-	-	84.7	77.2	85.5
	-	√	√	-	-	-	85.5	82.3	84.5
	√	-	√	-	-	-	84.4	82.3	85.8
	√	√	-	-	-	-	84.2	68.4	88.9
	√	√	√	-	-	-	91.7	77.2	81.8
	√	√	√	√	-	-	92.3	87.3	87.4
	√	√	√	-	√	-	94.2	89.1	92.7
	√	√	√	√	√	√	96.0	90.9	94.5

NOTE: ST=Swin-Transformer; CTR=Cosine learning rate; TL=Transfer Learning; P=Precision; R=Recall.

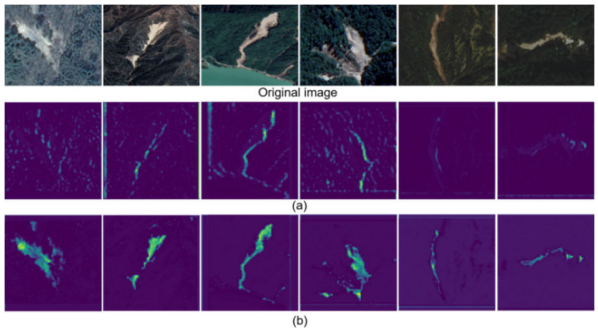


Fig. 10. Heatmaps of YOLOv5. (a) STAFM-YOLO and (b) Luding landslide dataset.

Table II shows the experimental results of STAFM-YOLO on the Luding landslide dataset, further validating that the various optimization measures significantly improved precision, recall, and mAP@50. After incorporating Swin-Transformer, CBAM, and BiFPN, precision increased by 21.6%, recall by 9.6%, and mAP@50 by 18.0%. Fig. 10 presents the comparative heatmaps of YOLOv5 and STAFM-YOLO on the Luding landslide dataset. The results indicate that STAFM-YOLO exhibits a higher focus on target regions and less interference in nontarget areas than YOLOv5. Particularly, noteworthy is its ability to effectively concentrate on positive sample regions across various object sizes, thereby reducing irrelevant environmental noise. These findings underscore the effectiveness of integrating the CBAM mechanism in enhancing network feature representation and the significant performance improvement achieved by employing the BiFPN structure for multiscale remote sensing object detection tasks. Adding the cosine learning rate and AdamW optimizer boosted precision by 0.7% and 2.7%, recall by 13.1% and 15.4%, and mAP@50 by 6.8% and 13.3%, respectively. Applying transfer learning, combined with Swin-Transformer, multiscale feature fusion, cosine learning rate, and AdamW optimizer, precision improved by 27.3%, recall by 29.1%, and mAP@50 by 36.3%. The results demonstrated that these improvements significantly enhanced the detection performance of STAFM-YOLO on different landslide datasets. This validates the effectiveness of the enhancement strategies and highlights their potential for broader applications.

Table III presents a detailed comparison of Cotnet-Transformer, PvT-Transformer, and C3-Swin-Transformer

TABLE III
RESULTS FOR DIFFERENT TRANSFORMER MODULES

Data	Module	Precision (%)	Recall (%)	mAP@0.5 (%)
Jiuzhaigou	Cotnet-transformer	88.3	86.9	89.7
	PvT-TransFormer	92.3	87.3	87.7
	C3- Swin-Transformer	95.7	89.9	90.5
Luding	Cotnet-transformer	91.1	92.7	91.3
	PvT-TransFormer	94.4	92.7	91.8
	C3- Swin-Transformer	96.0	90.9	94.5

TABLE IV
RESULTS FOR DATA ENHANCEMENT STRATEGY

Data	Programs	Precision (%)	Recall (%)	mAP@0.5 (%)
Jiuzhaigou	No data enhancement	90.5	84.2	86.6
	Data enhancement	95.7	89.9	90.5
Luding	No data enhancement	88.3	86.9	89.7
	Data enhancement	96.0	90.9	94.5

modules. On the Jiuzhaigou dataset, the C3-Swin-Transformer achieved the highest performance across all metrics, with a precision of 95.7%, recall of 89.9%, and mAP@0.5 of 90.5%. The PvT-Transformer showed competitive recall (87.3%) but fell short in precision (92.3%) and mAP@0.5 (87.7%) compared to the C3-Swin-Transformer. Among the three modules, the Cotnet-Transformer recorded the lowest precision (88.3%) and recall (86.9%), although its mAP@0.5 reached 89.7%. On the Luding dataset, the C3-Swin-Transformer once again outperformed the other models, achieving a precision of 96.0%, recall of 90.9%, and mAP@0.5 of 94.5%. The PvT-Transformer attained a high precision of 94.4% and identical recall (92.7%) to the Cotnet-Transformer but delivered a slightly higher mAP@0.5 (91.8%). In contrast, the Cotnet-Transformer achieved a precision of 91.1%, recall of 92.7%, and mAP@0.5 of 91.3%. Overall, the C3-Swin-Transformer demonstrated superior performance across both datasets, consistently achieving higher precision and mAP@0.5 scores compared to the other modules, particularly excelling in complex scenarios. The PvT-Transformer delivered reliable performance but had slightly lower mAP@0.5 scores, while the Cotnet-Transformer, although competitive, lagged in precision and recall metrics.

The impact of data enhancement on model performance is summarized in Table IV. In the Jiuzhaigou dataset, the implementation of data enhancement significantly improved the model's performance, with precision increasing from 90.5% to 95.7%, recall from 84.2% to 89.9%, and mAP@0.5 from 86.6% to 90.5%. Similarly, on the Luding dataset, data enhancement led to notable improvements, raising precision from 88.3% to 96.0%, recall from 86.9% to 90.9%, and mAP@0.5 from 89.7% to 94.5%. These results demonstrate that the data enhancement strategy effectively boosts model accuracy and robustness across different datasets, particularly in challenging environments.

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TABLE V
DETECTION RESULTS OF DIFFERENT MODELS ON THE JIUZHAIGOU DATASET

Model	Precision (%)	Recall (%)	mAP@0.5 (%)	FPS (ms)
STAFM-YOLO	95.7	89.9	90.5	6.61
Yolo-tiny	84.3	81.9	85.6	168.48
Yoloxl	83.5	89.2	81.7	7.83
Yoloxm	87.3	79.8	79.1	8.93
Yoloxs	84.6	76.8	79.0	7.67
Yolov2-r50-tiny	86.5	75.6	77.5	65.70
Yolov3	84.1	79.3	71.5	6.61
Yolov3-tiny	89.1	71.9	73.0	2.76
Yolov3-r34	70.2	68.3	64.3	34.52
Yolov3-mobilenet	79.1	70.6	75.9	13.29
Yolov4s	92.1	89.1	89.0	9.57
Yolov4l	90.4	85.4	88.9	12.2
Yolov4m	91.1	92.7	91.3	10.6
Ssd-vgg16	89.7	83.2	86.4	60.97
Faster-rcnn-r50	87.2	73.6	74.8	21.55
Faster-rcnn-r101	85.1	77.6	77.2	18.53
Faster-rcnn-swin	86.3	78.9	77.8	16.16

TABLE VI
DETECTION RESULTS OF DIFFERENT MODELS ON THE LUDING DATASET

Model	Precision (%)	Recall (%)	mAP@0.5 (%)	FPS (ms)
STAFM-YOLO	96.0	90.9	94.5	12.2
Yolo-tiny	85.2	78.6	85.6	168.48
Yoloxl	88.3	71.2	81.7	7.83
Yoloxm	87.5	70.6	79.1	8.93
Yoloxs	88.6	73.1	79.0	7.67
Yolov2-r50-tiny	85.2	75.6	77.5	65.70
Yolov3	82.4	82.5	83.7	9.74
Yolov3-tiny	92.3	87.3	88.5	3.63
Yolov3-r34	72.3	67.5	64.3	34.52
Yolov3-mobilenet	79.6	70.2	75.9	13.29
Yolov4s	85.4	82.5	83.0	11.2
Yolov4l	88.2	78.9	80.7	15.1
Yolov4m	80.3	73.7	75.7	12.5
Ssd-vgg16	84.2	83.6	86.4	60.97
Faster-rcnn-r50	79.6	73.1	74.8	21.55
Faster-rcnn-r101	85.2	77.3	77.2	18.53
Faster-rcnn-swin	86.3	78.1	77.8	16.16

B. Comparison With Other Models

To further evaluate the performance of the STAFM-YOLO model, a comparison was conducted with 16 commonly used object detection models, including Yolo-tiny, Yoloxl, Yoloxm, Yoloxs, Yolov2-r50-tiny, Yolov3, Yolov3-tiny, Yolov3-r34, Yolov3-mobilenet, Yolov4s, Yolov4l, Yolov4m, Ssd-vgg16, Faster-rcnn-r50, Faster-rcnn-r101, and Faster-rcnn-swin. These models were trained on the Jiuzhaigou and Luding landslide datasets. Evaluation metrics included precision, recall, mAP@0.5, and FPS.

Table V presents the experimental results of these models on the Jiuzhaigou landslide dataset. The results indicate that the STAFM-YOLO model achieved a precision of 95.7, recall of 89.9, and mAP@0.5 of 90.5. Compared to other models, STAFM-YOLO showed a significant improvement in overall performance. The precision value surpassed all other models, and the recall and mAP@0.5 also showed considerable enhancement, only slightly lower than Yolov4m. In addition, STAFM-YOLO demonstrated excellent computational efficiency, with an inference speed of 6.61 ms per image and an FPS higher than the other models, except Yolov3-tiny.

Table VI illustrates the experimental results of the Luding landslide dataset. The STAFM-YOLO model achieved a precision of 96.0, recall of 90.9, and mAP@0.5 of 94.5. In contrast, other models' precision, recall, and mAP showed a decline. Moreover, the STAFM-YOLO model's FPS was 12.2, marginally higher than the other models, except Yoloxl, Yoloxm, Yoloxs, Yolov3, Yolov3-tiny, and Yolov4s.

C. Visible Results

Figs. 11 and 12 illustrate some detection results of STAFM-YOLO on the Jiuzhaigou and Luding datasets. The model accurately identified isolated landslides in all correctly detected regions. When historical landslide areas were partially covered by vegetation, STAFM-YOLO exhibited high recognition capability. This is because transfer learning addresses the issue of small sample sizes, while Swin-Transformer enhances feature extraction capabilities, meeting the requirements for high-precision computation. The CBAM and BiFPN modules enable the fusion of nonuniform contextual information. The AdamW optimizer and cosine learning rate schedule effectively prevent overfitting, accelerate network convergence, and facilitate the rapid identification of earthquake-induced landslides. In addition, the model accurately ignored these nontarget regions in areas with early soil and rock, strip-like landslides, and roads located on the right side of the landslide area. This indicates that STAFM-YOLO effectively distinguished between relevant landslide features and irrelevant environmental information. The nontarget regions, such as roads and early soil, did not contribute to landslide characteristics, and their exclusion helped the model focus on the critical features for accurate detection. For targets of varying sizes, STAFM-YOLO effectively focused on the positive sample areas and minimized interference from irrelevant information. Even in complex terrains with two or more landslides, the model effectively identified the boundary of each landslide and detected all landslides accurately.

VI. DISCUSSION

The primary challenge in detecting earthquake-induced landslides using remote sensing images lies in the significant variation in landslide shapes and sizes and the scarcity of samples. General object detection models struggle to achieve a balance between high accuracy, computational capability, and small sample size, which remains challenging. This study proposes the STAFM-YOLO algorithm based on transfer learning, which combines the C3-Swin-Transformer, CBAM, and BiFPN structure. Furthermore, the network is trained using AdamW optimizer and cosine learning rate strategy, accelerating model convergence, and enhancing training speed.

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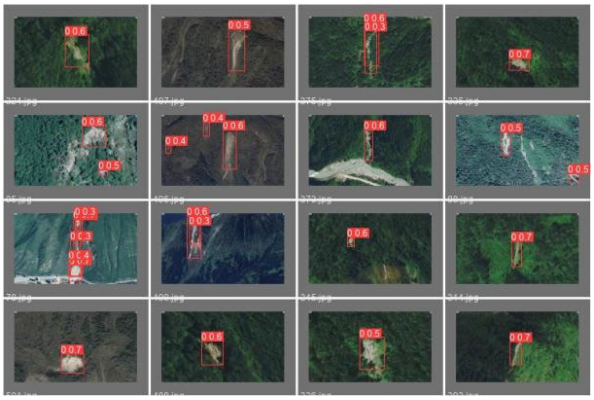


Fig. 11. Target detection of Jiuzhaigou landslide detection results of STAFM-YOLO model.



Fig. 12. Target detection of Luding landslide detection results of STAFM-YOLO model.

The algorithm enhances high-level features by integrating multiple low-level features to increase feature dimensions and information. STAFM-YOLO outperforms other algorithms in detecting small targets in remote sensing images of landslides.

Based on the YOLOv5 model, adding the Swin-Transformer module increased the mAP@50 of the Jiuzhaigou dataset from 67.0 to 81.9 and the Luding dataset from 69.3 to 86.4, significantly enhancing the model's information extraction capability. Further incorporating the CBAM and BiFPN structures increased the mAP@50 of the Jiuzhaigou dataset by 6.3% and the Luding dataset by 1.6%, demonstrating their effectiveness in multifeature fusion. Among Cotnet-Transformer, PvT-Transformer, and C3-Swin-Transformer, the C3-Swin-Transformer achieved the best performance on the Jiuzhaigou dataset, with precision of 95.7%, recall of 89.9%, and mAP@0.5 of 90.5%. Data enhancement significantly improved performance: precision, recall, and mAP@0.5 rose from 90.5%, 84.2%, and 86.6% to 95.7%, 89.9%, and 90.5% on the Jiuzhaigou dataset, and from 88.3%, 86.9%, and 89.7% to 96.0%, 90.9%, and 94.5% on the Luding dataset. On this basis, introducing the AdamW optimizer and cosine learning rate strategy improved the mAP@50 of the Jiuzhaigou dataset by 1.5% and 0.6%, respectively, and the Luding dataset by 0.7% and 5.3%, respectively, indicating the positive impact of these strategies on model performance. Compared with other common models, the STAFM-YOLO model achieved the best mAP@50 values while maintaining low FPS, demonstrating its ability to balance accuracy and efficiency with small sample sizes.

By integrating advanced modules and optimization strategies, the precision and recall rates are improved. This means that the model can more accurately identify potential landslide areas from aerial or satellite imagery, thereby reducing the risk of misclassification and improving early warning capabilities. For example, in remote areas such as Jiuzhaigou and Luding, which are difficult to monitor manually, the STAFM-YOLO model's ability to efficiently process images from multiple sources can help to identify landslide areas faster and more accurately. This ultimately contributes to more effective disaster preparedness and relief. In addition, the model's low FPS ensures that it operates on devices with limited computational resources, such as edge devices deployed at field monitoring stations. This adaptability enhances the practical applicability of the model in resource-limited environments, enabling continuous real-time monitoring in high-risk areas.

Despite significantly reducing data requirements through transfer learning, STAFM-YOLO still requires a fully annotated landslide dataset for training. Specifically, a detailed and accurate annotation of landslide boundaries and characteristics is essential to ensure reliable model training and validation. This task remains costly and time-consuming, requiring expertise in remote sensing and landslide analysis. To mitigate these challenges, we plan to explore unsupervised learning techniques in future work, aiming to identify and label unannotated earthquake-induced landslide data more efficiently.

VII. Conclusion

This study proposes the STAFM-YOLO model for remote sensing earthquake landslide image detection, leveraging attention mechanisms and multiscale feature fusion. The integration of the C3-Swin-Transformer module enhances feature extraction, meeting high-precision requirements. The CBAM mechanism in the feature fusion layer boosts feature representation, while the BiFPN structure facilitates efficient bidirectional cross-scale feature fusion, capturing multiscale features, and enabling the fusion of nonuniform contextual information. The use of the AdamW optimizer and cosine learning rate strategy accelerates model convergence and improves training speed. In addition, transfer learning effectively addresses challenges posed by small sample sizes. Experimental results on the Jiuzhaigou and Luding landslide datasets demonstrate the algorithm's efficacy in detecting earthquake-induced landslides under small sample conditions. Future research will focus on enhancing manual annotation efficiency by exploring methods to speed up data processing and reduce expert intervention. In addition, unsupervised learning techniques will be employed to accurately detect small-scale landslides.

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DATA AVAILABILITY

The code of this research is now published in https://github.com/mengshaoqiang/A-Transfer-Learning-Network-for-Earthquake-induced-Landslide-Remote-Sensing-Image-Detection. The open remote sensing datasets that support the results of this research are available online. The VOC dataset is now available at http://host.robots.ox.ac.U.K./pascal/VOC/. The Bijie dataset is available at http://gpcv.whu.edu.cn/data/Bijie_pages.html.

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